Multiple-choice answers 1-4

1. Study design (C)

A: More observations reduce chance variation, not systematic selection bias.

B: Random allocation of a treatment defines a randomised experiment.

C - correct: Sleep is observed. Selection bias and confounding remain possible.

D: A common measurement does not change how students were recruited.

2. Linking student records (D)

A: Removing names alone does not resolve consent or re-identification risks.

B: Replacing an identifier does not necessarily make linked records anonymous.

C: More data can increase the risk of identifying an individual.

D - correct: Check the new use, minimise the linked data and protect access.

3. Transforming variables (A)

A - correct: $10 - 2 \times 7 = -4$. SD is multiplied by 2 and correlation changes sign.

B: Multiplication by -2 doubles the SD.

C: Reversing the direction of Sleep changes the correlation's sign.

D: The transformed mean is $10 - 14 = -4$.

Mean = 10 - 2 x 7 = -4; SD = 2 x 1.2 = 2.4; correlation = -0.4114.

4. Normal probabilities in R (D)

A: The third argument of pnorm is the SD, not the variance.

B: The subtraction is reversed, giving a negative number.

C: This adds the two tails outside the required interval.

D - correct: Subtract the lower-tail area at 5.8 from the area at 9.4.

The probability is 0.8185946141. The 68-95 rule gives approximately 0.815.

Multiple-choice answers 5-8

5. Measurement error (C)

A: Averaging reduces chance error. The common +20 minute bias remains.

B: The spread of the average is smaller than the spread of one reading.

C - correct: Bias remains +20. The SE is 12 / sqrt(36) = 2 minutes.

D: The fixed bias remains. SE is divided by sqrt(36), not 36.

6. Conditional probability (B)

A: This conditions on 2+ drinks rather than short sleep.

B - correct: The relevant group contains 80 students, of whom 44 report 2+ drinks.

C: This is the joint probability among all 160 students.

D: This is the marginal probability of 2+ drinks.

7. Binomial probability in R (A)

A - correct: At least two is the complement of zero or one.

B: This gives more than two, so it excludes X = 2.

C: This gives exactly two.

D: This gives at most two.

8. Sampling distributions (D)

A: SE scales with 1/sqrt(n), not 1/n.

B: A larger sample reduces the SE.

C: The EV of a sample mean remains the population mean.

D - correct: SE=1.2/sqrt(100)=0.12. Four times n gives half the SE.

Multiple-choice answers 9-13

9. Bootstrapping the mean (B)

A: Without replacement, every value appears once. Each mean is unchanged.

B - correct: Each bootstrap sample has the original size and resamples the observed values with replacement.

C: This simulates from a Normal model rather than resampling the observed data.

D: 10,000 is the number of replicates. Each resample should still contain 160 values.

10. Using a confidence interval (B)

A: A sample estimate can differ from the null through chance variation.

B - correct: A matching two-sided 5% test retains null values inside the 95% CI.

C: The printed p-value tests $\mu D = 0$. The new p-value is 0.3332.

D: The parameter is fixed. 95% refers to the CI method's long-run coverage.

SE = (0.5 - 0)/2 = 0.25. New t = (0.5 - 0.75)/0.25 = -1, with 15 df. Two-sided p = 0.3331701.

11. Linear regression (A)

A - correct: Increasing spread suggests heteroscedasticity. The usual test needs caution.

B: The slope test does not check the constant-variance condition.

C: Sleep was observed. Confounding remains possible.

D: A QQ plot checks Normality, not every LINE condition.

12. Choosing the chi-squared test (C)

A: Equal-proportion GOF concerns one variable, not the association of two.

B: 44 is observed. Expected counts are calculated under independence.

C - correct: E=80 x 64/160=32 and df=(2-1)(3-1)=2.

D: rc-1 is not the df formula for an independence test.

E = 80 x 64 / 160 = 32; df = (2 - 1)(3 - 1) = 2. All expected counts are at least 20.

13. Goodness of fit (D)

A: Unequal counts can arise through chance even when the model holds.

B: Retaining a model does not prove it.

C: The p-value assumes the null model. It does not give its probability.

D - correct: $p > 0.05$, so the data do not provide sufficient evidence against the model at this level.

Short answers 1 and 2

1. Study limitations (4 marks)

Sleep was not randomly assigned. Workload or prior preparation could be associated with both sleep and quiz scores, so the fitted slope alone cannot establish a causal effect. The volunteers were reached at evening study spaces and may differ from other first-year students, so the result cannot automatically be generalised to all first-year students.

1 mark each:
1. Identify the causal overstatement.
2. Explain the lack of random allocation or a plausible confounder affecting both sleep and score.
3. Identify the unsupported population generalisation.
4. Link that limit to volunteers recruited at evening study spaces. Equivalent well-supported wording earns credit.

2. P-value interpretation (4 marks)

If sleep group and caffeine category were independent, the probability of obtaining a chi-squared statistic of 16.543 or larger would be approximately 0.0002557, assuming the test conditions hold. Since $p < 0.05$, reject the null of independence. The data provide evidence of an association. This does not establish causation.

1 mark each:
1. Condition the probability on independence being true.
2. Specify a chi-squared statistic of 16.543 or larger.
3. Compare p with 0.05 and reject the null.
4. State evidence of association between sleep group and caffeine category. The p-value does not give the probability of the null hypothesis.

HATPC working

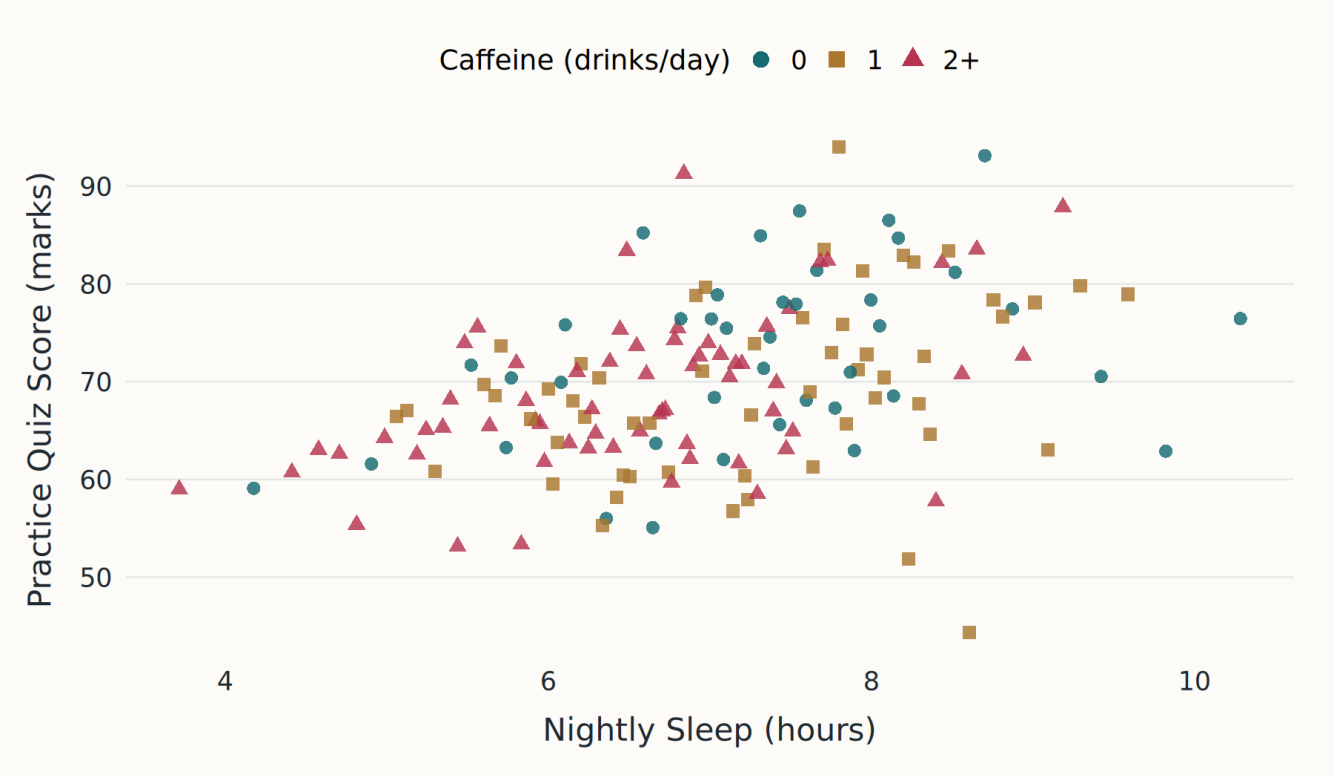
H: independence versus association. **A:** independent student records and adequate expected counts.
T: Expected rows are (20, 28, 32). Sum $(O - E)^2/E = 2 \times (3.2 + 0.5714286 + 4.5) = 16.542857$. df = 2.
P: Upper-tail area = 0.0002557197.
C: Reject at 5%, with evidence of association under the model assumptions. Recruitment still limits population generalisation.

Short answer 3

A question, a plot and code (5 marks)

Question: How does the association between sleep and quiz score vary across caffeine categories?

```
ggplot(students,
aes(x = Sleep, y = Score,
colour = Caffeine, shape = Caffeine)) +
geom_point() +
labs(x = "Nightly Sleep (hours)",
y = "Practice Quiz Score (marks)")
```



1 mark each:
1. A research question involving all three variables.
2. A scatterplot with labelled Sleep and Score axes.
3. A clear key for caffeine categories.
4. Correct data frame and x/y mapping in ggplot().
5. geom_point() with colour or shape mapped to Caffeine. A labelled schematic is sufficient.
Equivalent coherent questions and graphical mappings earn credit.

The displayed figure adds a presentation theme and an accessible colour/shape scale to the basic answer code. A multiple-regression or interaction test is not required.

Course sources and revision

Question alignment

Q1: Topic 1 pp. 25–38; Topic 8 pp. 12–22; Challenge 1 and 8; sample MCQ Q2, Q14; sample EA part (a).

Q2: Module 1, Topic 1: ethics and data linkage; current sample MCQ Q1. LO1.

Q3: Topics 3 and 5; Challenges 3.3 and 5.3; Exam Guide transformations; sample MCQ Q4.

Q4: Topic 4 pp. 10–25; Challenge 4; Revision Quiz 4 Q18/Q25/Q26; sample MCQ Q6.

Q5: Module 2, Topic 4: measurement error; Module 3, Topic 7: SE of a mean; current sample MCQ Q7. LO4 and LO6.

Q6: Topic 6 conditional probability; Challenge 6 prosecutor fallacy; Revision Quiz 6 Q1/Q7/Q8; Week 13 recap p24.

Q7: Module 3, Topic 6: Binomial model and pbinom; Challenge 6; current sample MCQ Q12 and Exam Guide. LO6.

Q8: Topic 7 pp. 26–32,45,55; Challenge 7 and 8; Revision Quiz 7 Q6/Q10; Week 13 workshop Report 2 Q2.

Q9: Module 3, Topic 8: bootstrapping procedure, lecture pp. 48–52. LO6.

Q10: Topic 10; Revision Quiz 10 Q1; sample MCQ Q17; Week 13 workshop Report 2 whiteboard task.

Q11: Topic 5 pp. 39–50; Topic 11 pp. 47–53; Challenge 5 residual audit; sample MCQ Q8/Q10/Q20.

Q12: Topic 11 pp. 6–7,23–32; Challenge 11; Revision Quiz 11 Q8/Q11/Q19; sample EA Source 3.

Q13: Module 4, Topic 11: chi-squared goodness of fit; Challenge 11; current sample MCQ Q18, Q29–30. LO7 and LO8.

The short answers follow the criticism, p-value interpretation and graphical-code tasks in the official sample extended answer. These are original practice questions with suggested marks, not an official marking scheme.

Official resources

Canvas course 74355: Topics 1-11, weekly challenges, revision quizzes, Week 13 revision workshop, Exam Guide and current sample papers. The written-source review was completed on 3 November 2026. Full source links and the question audit are in the project documentation.

This revision reopened the current Canvas learning map, exam-preparation page, sample MCQ paper, extended-answer sample, Topic 8 lecture and Challenges 4, 6, 8 and 11. It also used the original complete written-source inventory. Recordings and separately linked interview transcripts were not reviewed. No separate official extended-answer marking rubric was linked.

Try the weekly challenges independently, repeat the revision quizzes and complete the current sample under timed conditions. Keep an error log that explains the wrong choice and the reasoning to use next time. Revise the full core syllabus beyond this session.

More resources: bodhimcnally.com

Presentation and interactive examples

Presenting and live voting

Live votes: Extract the complete project. Configure a permanent voting server with python scripts/deploy_live_questions.py, then open Start_Live_Questions.bat or run python scripts/launch_live_questions.py. Students scan the joining QR and can join mid-session without an app or account.

Space/right arrow or A closes voting and reveals the explanations with A-D percentages. Q enlarges the joining QR. Late arrivals receive the current question. Revealed questions stay locked. Add --with-shiny to run the R widgets alongside polling.

Local integration and 100 simultaneous voter connections passed. Public Cloudflare connectivity is unverified because this build environment blocks tunnels. Temporary links have no uptime guarantee and a 200 in-flight request limit. Use the Worker setup for the event and check joining from a phone.

Offline: Open masterclass.html. The option selection, answer reveals, sample-size controls and null-value controls work without an internet connection. The final website and source links need internet access.

R Shiny: Extract the complete project. Install shiny and ggplot2, then run source("launch_shiny.R") from the project folder in RStudio. The companion app opens the same presentation with live R plots embedded on the sample-size and paired-test slides. Rscript launch_shiny.R also works; open the localhost address printed in the console.

The sampling widget calculates simulated sample means with R and overlays their theoretical Normal density. Changing the number of replicates smooths the approximation without changing theoretical SE. The paired widget reruns t.test(D, mu = selected value). The interval stays fixed while t and p change.

Error log

The slide uses an original Notion-style mockup. The editable mockup HTML and an importable CSV template are in resources/. The example entries cover reversed conditioning, the sign of correlation after a negative transformation and using output from the wrong null hypothesis.

MCQ advice sources

NBME (2024), Item-Writing Guide, pp. 12-13 and technical item flaws: the expected answer before the options and option-length cues. Kruger, Wirtz and Miller (2005), JPSP 88(5), 725-735, DOI 10.1037/0022-3514.88.5.725: the first-instinct fallacy. Butler and Roediger (2008), Memory & Cognition 36(3), 604-616, DOI 10.3758/MC.36.3.604: feedback after MCQ practice. The last-resort quotation is supplied by Bodhi and deliberately labelled facetious, without an invented attribution.

Outcome mapping

LO1-LO10 follow the supplied wording and official DATA1001 S2 2026 outline. The blueprint gives the full mappings. SA3 practises the LO10 reporting component. Teamwork, oral presentation and interrogation require separate assessment.